Question 1: Containerize a sample application using Docker: write a Dockerfile, build the image, and run the container. Document the image layers and container behavior.

Executed on Ubuntu 24.04 LTS (WSL2) by Meenakshi Rakesh. Repository: https://github.com/meenakshi16rp/Devops-lab-2

1. The Dockerfile

Dockerfile (committed at the repository root)

```
# ENSP461 Lab 2 - container image for the sample Flask app
FROM python:3.12-slim

ARG APP_VERSION=2.0.0
ENV APP_VERSION=${APP_VERSION}

WORKDIR /app

# install dependencies first so this layer is cached
COPY app/requirements.txt .
RUN pip install --no-cache-dir -r requirements.txt

# copy application code
COPY app/ .

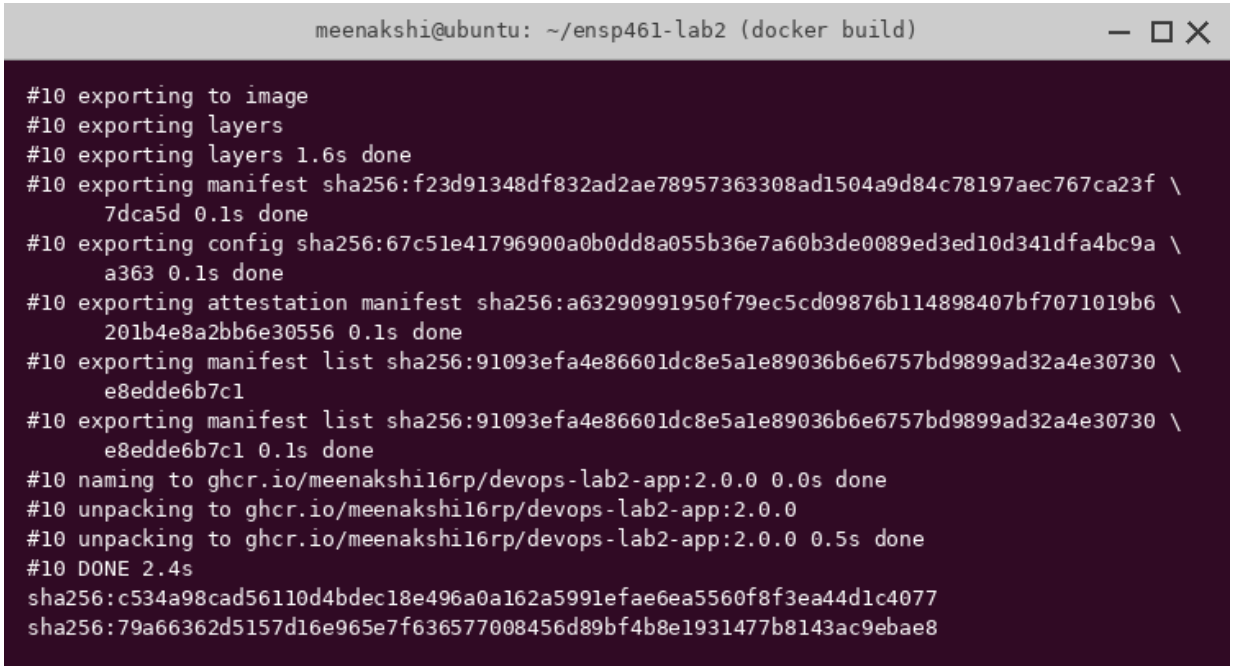
EXPOSE 5000

# run as a non-root user
RUN useradd -m appuser
USER appuser

CMD ["python3", "app.py"]
```

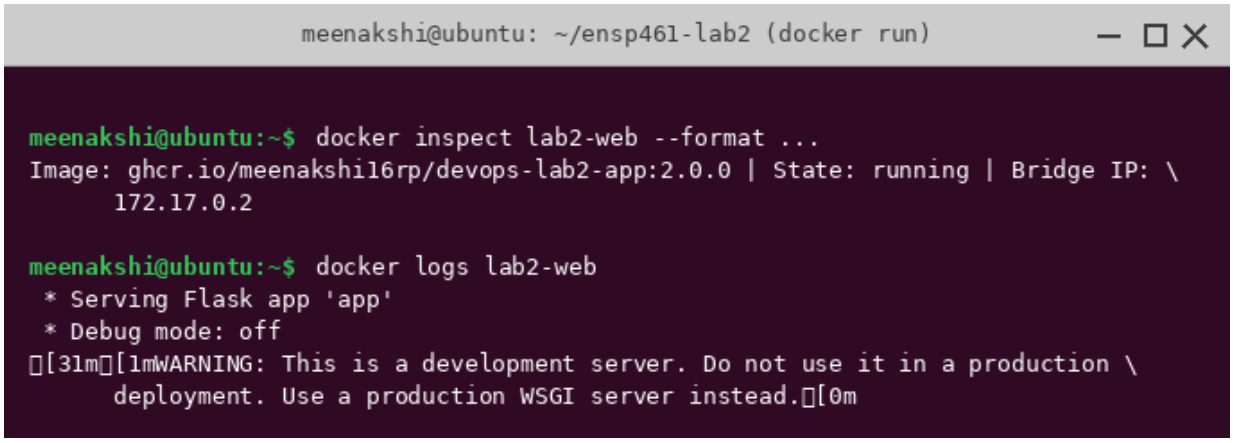
- Base image python:3.12-slim keeps the image small (207 MB disk usage, 50.3 MB content).
- requirements.txt is copied and installed before the source, so dependency layers stay cached when only the app code changes.

2. Building the image

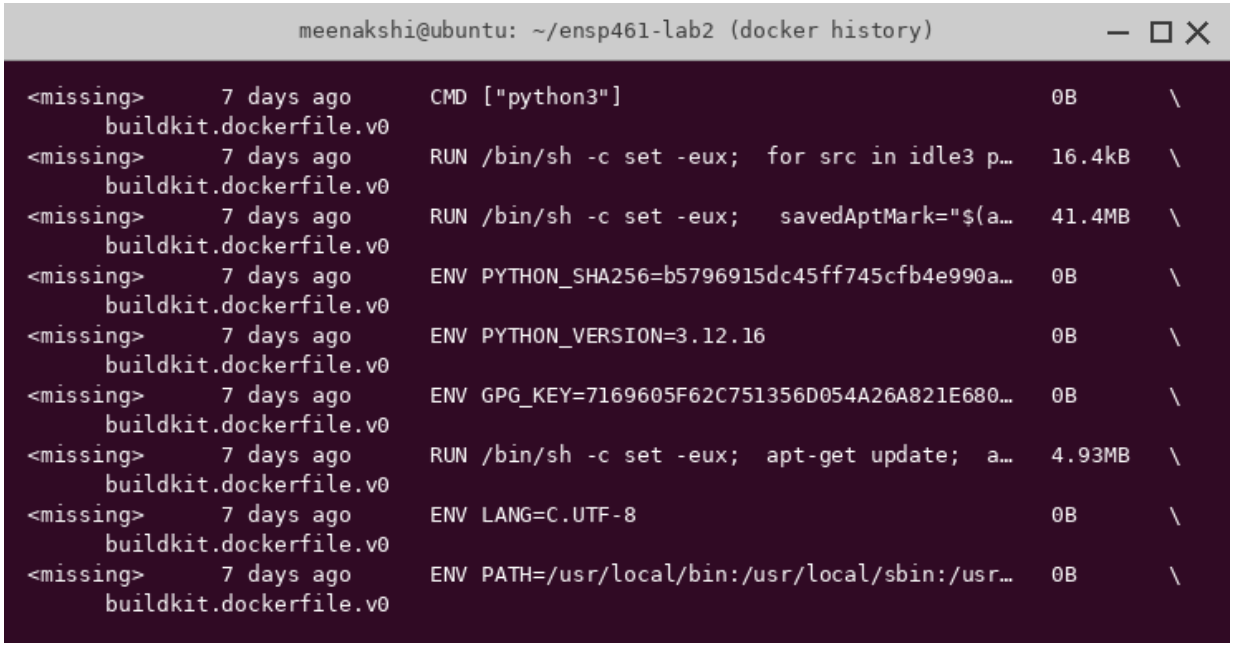



```
meenakshi@ubuntu:~$ docker images ghcr.io/meenakshi16rp/devops-lab2-app
REPOSITORY                              TAG       SIZE
ghcr.io/meenakshi16rp/devops-lab2-app   2.0.0     207MB
```

3. Running the container and verifying behavior



4. Image layers



- docker history shows each Dockerfile instruction as a layer: WORKDIR, COPY requirements (12.3 kB), pip install (22.6 MB), COPY source (20.5 kB), EXPOSE/ENV/CMD metadata layers.
- Container behavior: the Flask server binds 0.0.0.0:5000, / returns the greeting and version, /health returns UP, and /info reports the container hostname.

5. Observations

- The first run exited silently because app.py had no app.run() entry point - the container exited 0 with no logs; the fix was adding the entry block and rebuilding.
- A missing jsonify import then produced HTTP 500 with a NameError in docker logs - the console log pinpointed both defects quickly.
- Layer caching: rebuilding after changing only app/app.py reused the pip install layer, making the rebuild finish in seconds.